

ActInf GuestStream #033.1 ~ "A taxonomy of surprise definitions", Alireza

GuestStream | Alireza Modirshanechi | 1:39:21

hello and welcome everyone it's active
guest stream

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we are here with Ali and alireza modir
shanichi we're going to hear about the
taxonomy of surprise definitions there
will first be a presentation followed by
a discussion

so Ali Reza thank you for joining please
take it away with the presentation we'll
be looking forward to this and the
discussion

thank you very much Daniel and Ali for
inviting me

um so as you said I'm going to talk
about a taxonomy of surprise definitions
uh which is um basically all the
materials or most of the materials that
we publish in this recent paper in the
Journal of mathematical psychology uh
that is a joint work with johanni Brea
and Wolfram gerstner and lufranker is my
PhD supervisor

so talking about surprise let's start
with a simple top experiment

um let's say that you want to plan
something for your weekend and to plan
something for the weekend we need
information like how the weather is
going to be or we can plan something
and or some cues like the spring that
are informative about the weather of the
weekend or that uh the weather forecast
says is going to be sunny
given these queues we can make some

predictions that oh weather forecast
said it's sunny and it's spring so most
likely it's going to be sunny
let's say that is Friday evening you
plan that for example go for a bike you
go to sleep wake up the day after uh
open the window and it's snowing and
seeing this note given that you expected
the sunny weather this mismatch is
what's been uh natural language would
agree on that it's a feeling of surprise
you know we can easily talk about this
moment say that yeah I was surprised
because I had planned to go for a bike
but then in scientific Community we can
ask this question that for these moments
that you see a snowy weather when
opening the window what does really
happen in our brain
and uh

lots of literature in a long lasting
debate in neuroscience and psychology
about different roles of surprise in the
brain and there's this slew of different
uh what I call surprise related signal
things like prediction error free energy
surprisal or Bayesian surprise and
people say that these things are related
to learning play important role in model
building or predictive coding some
people say that they're quite important
for exploration for example surprising
even attract attention or drive
curiosity
uh there are also literature and memory
that surprising events um
segments or continuous stream of
observation so they help us to segment

our memory and of course there are lots of work and physiological signals of uh surprise signals for example when there is something surprising your pupil dilates or EG signal has a peak and so on so forth

the thing is that if I enable look at this picture and from far away it seems like that surprises everywhere in the brain

but then the puzzle is that it's really the surprise in the sentence surprise martial synaptic elasticity same as the one in the sentence surprise Drive curiosity or not

that's basically the motivation of the main motivation of our work that uh different studies in neuroscience and psychology use similar words but to refer to different aspects of the moment of surprise

and I would call this art piece the moment of surprise and uh basically the thing is that different studies different experiment different paper they all talk about moments like these situations like this that we can all agree on to call it a surprising moment but they talk about different aspects of it

the way that some we summarize it or put it in verb in words in the abstract of the paper was there is no consensus on the definition of surprise when I posted the paper online on Twitter the linguist margin has to laugh wrote in reply that it's unsurprising that there is no consensus on the

definition of surprise because it's not a technical term everyday words are usually somewhat awake and uniform definitions are desirable only for technical terms of science and I cannot agree more with that and I want to emphasize that today I'm not going to talk about the meaning of surprise when we talk over a coffee break it's in the context of neuroscience and psychology and when we talk about these moments and the role of these surprise related signals in the brain so people are going to details and math and all the framework um I want to present the paper in a nutshell that what is the Absurd of the paper what we do is that we take many papers from the field and we identify several definition of surprise at least 10 of the definitions here in the paper that are 18 with all the details and these definitions include things like ansar reward prediction error confidence corrected surprise Bayesian surprise positive surprise and many things that people really use in different contexts then we ask this question that what are their similarity and differences between these different surprise definition and we have three main results that is uh analyzing these different definitions from different aspects well first main result is a technical classification so to talk about the technical classification I would like to introduce a short notation

we call observation y we call Q 's X
things that are going to be informative
about the observation
and then predictions can be seen as
either a probability distribution over
the observation given the cues or one
estimate of the next observation
and looking at this uh surprise
definition we could
um identify a category of these
definitions that what we call
observation mismatch to price definition
they only depend on the next estimate so
they don't care that okay what does the
probability of weather being sunny or
rainy or snowy they say that I predict
that the weather is going to be sunny
and now it's not sunny and what is the
difference between being sunny and being
snowy and that's the surprise there is
another group that we call probabilistic
mismatch they actually look at this
distribution and say that oh
uh it's true that I didn't expect the
snowy weather but it was not too
unlikely so it may be not as surprising
as I would expect it so they work with
the with a distribution instead of a
single point estimate or predictor
um
actually this uh conditional
distribution is a marginalization of
something bigger
because there is actually some
uncertainty about what how accurate for
example weather forecast is about the
weather of the next day and that what we
would call I would put in this parameter

Theta that are underlying rules of the environment or hypothesis the reliability of the weather forecast and we don't have access to the true value of it so we make it believe and this P_i of theta is our belief it's a distribution over these possibilities and actually the marginal distribution of next observation given the queue is this marginal probability and the third group of the surprise definitions people defining these definitions uh argue that actually what we care about is this belief that for example how much we learn how much we change this belief uh by observing a new new observation and those are the definitions that we call belief mismatched surprise definitions so this was a technical classification we put different definitions in three different groups and say that there is some uh difference in how they depend on uh the belief of subject then we go to a conceptual labeling and that's what we call the taxonomy and we talk about what is the concept or what is the conceptual arguments behind the definition of each of each of these surprise definition one group uh that's what we call prediction surprise so they uh they care about how accurate or inaccurate the prediction is how expected or unexpected the next observation is the more accurate the less surprising the observation then there's another category that we call change Point

detection surprise
they don't only care about the
prediction they always compare
prediction with a baseline prediction
if something is unexpected but that
thing is unexpected under any hypothesis
then they claim that is not surprising
because it would be just kind of uh an
outlier so I should not consider it as
surprising and these are important to
detect change points
um because
what they care about is to see if there
is another hypothesis under which the
current observation would be more likely
for example in this case of weather
forecast during covid time there was
this argument that there is this paper
I'm citing here that because there is
much less flight uh going on we had way
less data and sensors so weather
forecast suddenly stopped being accurate
and actually if you were checking
weather forecasts you could see uh
without even knowing the news that uh it
gets less and less accurate as time pass
on and this kind of change Point
detection surprise definitions are
somehow optimal I show later on that
they're optimal for detecting these kind
of unobserved changes
uh the third group or the third class
are what we call Information Gain
surprise so no matter if there is a
change or not uh this new observation
made me learn something more about those
underlying rules of the environment so
uh which

Media or which weather forecast is more reliable than the others every day that I see the weather I would use that to update my belief about different weather forecasts

and the last category is no matter which of these ones I care about what what is the concept behind the what is the conceptual argument behind the definition the argument is that if I'm more confident if I'm more confident about my prediction I should feel more surprised so the argument is that no matter what is the definition I should have an explicit term for confidence in my definition

so having this technical classification and this conceptual labeling we can arrange this list of surprise definition in this different boxes and have some structure and as you can see some of these boxes are empty and I would like to say

you know we have these mathematical definitions on the side but their justification come from the relation to experience uh to uh experiments and uh what they tell us about the brain so it can be the case that in future we see that none of these definitions are helpful or we see in some situation that we need another definitions and this uh figure here tell us that okay there are actually a

places that we can come up with new definitions in new categories but there are also restriction for example talking about Information Gain

if we want to talk about Information
Gain we should always talk about the
belief mismatch we cannot talk about
only at the level of observation or
marginal distribution because we are
talking about learning so we should talk
about these very high level uh hidden
variables or when we talk about
confidence we inevitably we need to talk
about probability distribution and we
cannot talk about Point estimates but uh
still we can talk about these three
different uh empty boxes here
and or last main result is although
these different definitions are in
different boxes but there are some links
between them so we found conditions
under which these definitions are
indistinguishable so there are some
situation that they're wanted on mapping
uh there is one-to-one mapping between
some of these definitions we cannot
really experimentally distinguish them
so um that was a summary of our work and
now I'm going to go uh to the
mathematical framework and go one for
one through all of these definitions and
they're linked to each other and please
let me know if there is any um
clarification question
so for mathematical framework
Let's uh look at one very traditional
experiment experiments for studying
surprise in the field of Neuroscience
it's what what's called volatile audible
task
uh in a typical output task at each time
point we have two possibility for uh the

stimulus it's either for example here a blue disc or a red square and for a while the blue disc is much less likely to appear on the screen than the other one and the reasoning behind the design of this experiment is that the participants who is watching this sequence on the screen one after another expect to see a the red screen more often than the blue disc and at the moment that it sees uh he or she sees the blue disc it's uh it's surprising and that would give them a feeling of surprise and a question that one can ask is that can we really quantify how surprised a participant feels at time t equal to four so if we know that this is the sequence until time t equal to 4 can be Quantified the amount of surprise or not and to talk about the feeling of participant about surprise of a surprise of an observation we should make some assumption about how participants perceive this sequence and we take it very common Assumption of the field that participants perceive their sensory observation as probabilistic outcomes of a generative model with hidden variables so at each time each of these observation come from some distribution with some hidden variable for example the way that participants we assume in a way that the way that participants think about the sequence of observation is that at each time with some probability uh one of

these uh stimuli appear on the screen
and over time given the sequence they
try to find this probability
uh to formally Define this we consider a
generative model

let's say that at time T we show the
observation what appear on the screen by
 Y_T and there is this hidden variable
which we call environment parameters
which is for example the probability of
blue disk appearing on the screen
and now this link for people who are
familiar with and going to make a
Bayesian Network and this link shows
that the environment parameter determine
the distribution over the observation
and I can repeat the same thing for time
 t plus one

something that I didn't talk about in
these volatile Oddball task is that for
a while we had this uh one image after
another with some fixed probability but
once in a while we have a hidden change
point which changed the probability of
blue disk and red square so previously
bluedisk was rare and after that the
other one can be rare or frequent
so the environment parameter over time
can change so we defined this parameter
uh this random variable change Point
indicator that at time t plus one if it
is one it means that there is a change
in the environment there is a new
parameter and if it is zero it basically
means that there is no change the
environment is stable and Θ_{t+1}
one is a copy paste of Θ_T So this
link just take that a t and take it to

$\theta + t + 1$

um so far the way that I present this in
uh this task uh was that different
observations are independent of each
other so at each time one of these
stimuli appear on the screen randomly
um this very beautiful paperless manual
and uh colleague in 2016 in Pillars
computational biology they argued it's
very nice they argue that even if the
sequence of observation is randomly uh
designed so this observation are
independent of each other
people when they are observing this
sequence assume that there's some
dependency so they what they uh what
they estimate over time is not really
probability of observing the blue disc
or Red Square they are estimating the
probability of observing blue disk after
Red Square or Red Square after Red
Square so what they assume is that
there's a link between y_t and y_{t+1}
one

so so far I have a model of this uh
Oddball task that says that each
observation depends on the environment
parameter and the previous observation
and there is a dynamic for this
environment parameter I do a bit of
trick just Define this dummy variable
which I call Q and for the Oddball task
I just copy paste the previous
observation in the queue but it uh the
reason that we use Q is to have a one
abstract notion of what is um what has
predictive power for the for predicting
next observation and for the for the

task with the action action can go to Q
for the example of the weather forecast
weather forecast can go to the queue and
all these things

and um now having this Motif I can
repeat it and make the whole sequence
and given this generative model given
this mathematical model of the
experiment I can now mathematically
Define the surprise of observing variety
plus one so with this assumption we can
uh we can propose some quantification of
surprise of one stimuli

and uh what is force mentioning is that
this generative model here is
generalization of many other models in
the field which means that many of the
experiments already done to study
surprise can be modeled by this
generative model and a results hold for
that

okay

um before going forward this generative
model also account for these uh moment
of surprise can model that that the Q
was X is here the observation was the
weather of next day is here and Theta
was environment's parameter was how
reliable for example the weather
forecast is about the next observation
and the change point is whether
something happened in the environment
for example the pandemic and decreasing
the number of planes

okay

and now I'm going to make this
definition more formal and discuss the
Dynamics of each of these different

levels of in the generative model

so

I start with the observations

and we assume this is the probability of

at time $t + 1$ observing y condition

on Q being X and parameter being Θ

and basically this equation here means

that this probability is time

independent so as long as I know the Q

and the parameter it doesn't matter that

if it's time Step 1 or time step t

in all the time the same distribution

which we uh show by $p(y|x)$ given X

so basically any change in the

environment goes to the parameter and

the distribution is fixed

um for the Q variables we don't put any

constraint and for example

it can even be independent of anything

that's object assumed which is the for

example for weather forecast uh

you just turn on the TV and see what's

going to be the prediction for next

state

then we have the change points indicator

and their Dynamic is at each time

independently we ask whether there is a

change or not and $CT + 1$ and time t

$+ 1$ is a bernardly binary sample

with probability of PC which is the

chain what we call change Point

probability

so at each time point with probability

PC there is either a change or not

then um this is the

most important in a way part of the

model that

how this parameters change over time we

consider prior belief π_0 and say that at time one before the experiment starts participants have some uh assumption and some belief about what are the parameters Behind these uh generative process and uh that we show with π_0 and at time T if there is no change in the environment if c_{t+1} is equal to zero θ_{t+1} is just copy paste of θ_T but if there is a change in the environment we assume that θ_{t+1} is sampled again independently of past from this prior belief

so we have a stable environment with the same data until c_{t+1} is equal to one then we have a change in the environment and a parameters is uh resampled from these prior belief um

these equations that I wrote here is fully describe the joint distribution of c_1 to c_{t+1} θ_1 to T θ_t plus one and so on and so forth the whole generative model is described by these definitions but if you want to see the precise mathematical definition please look at the definition one in the paper

before going forward I would like to show some notation that uh whenever I write something like this

I shortened the vendors whenever there's no ambiguity I shortened with the notation by dropping the random variables that are uh the capital letters

um then I Define what I call observers
belief at time T if you remember P_i zero
I Define it as initial belief and now P_i
 T is the Observer belief at time T that
is given that I observe y_1 to Y_T with
 q 's X_1 to x_t what is my
knowledge about the current parameters
of the environment which basically can
be summarized in the posterior
probability of θ at T given the
previous cues and observation
um

which in this example is that given a
sequence of days and a prediction of the
weather forecast I can make a belief on
how reliable the weather forecast is at
time t

and at the end they Define this marginal
probability which is the probability of
 Y given X and now I dropped θ and I
put the knowledge here and integrate
over all possibility that if the world
is at time t

how my knowledge about the world
identify the relation between q and
observations

so this was uh the mathematical
framework and given this mathematical
framework now I can go to a formal
investigation of different definitions
and uh present or classification in
distinguishability condition and
taxonomy

what is surprise

having this model I can ask this
question that condition on the previous
observations y_1 to T and Q variables X_1
to t plus one these ones and these ones

how surprising is the next observation

YT plus one

we are at time t we are going to observe

YT plus one and this is now a valid

mathematical question in this framework

to make some prediction about YT plus

one I need information about Teta T and

x_{t+1} that is the Q and these are

the rules of the environment whatever I

know about the rules of the environment

is summarized in my belief in my uh

in whatever knowledge this uh

observation and previous cues gave me

about t

and uh given the belief and the queue I

can get rid of I can put away throw away

all the previous observation and say

that I only need the belief and the Q

and surprise is basically a function

that gets us arguments the belief the uh

the queue and the next observation and

give a real value back

one famous example is what people call

surprisal or uh Shannon's surprise the

history is so long it has been used for

so long that I cannot really

going to the first paper that use it but

you can look at the paper of barter in

2013 for a review

which basically

space IQ and an observation

surprise of uh this observation is equal

to minus logarithm of marginal

probability so the more likely this

observation the less surprising it is

there's also another way to define it

which

um because here I had the knowledge at

time t one can say that there was a possibility to have a change Point here and uh one can extend this probability and say that oh there was a when I'm at time T and I'm predicting the next observation I should also consider this possibility that the Vault may change in between and I should put some weights on this prior knowledge so the next observation comes with some probability from the current Volt or from the uh prior to terminal word that is recent

uh so there are two possibilities for these definitions and

um what I want to emphasize is that in volatile environments we are not sure which one we should take necessarily one may say that the this definition makes more sense because it's the full probability but there are experimental evidence in this paper of Nassau and colleague in 2010 in the Journal of Neuroscience

what they argue is that when people observe Y_T plus one they update their knowledge by assuming that there might have been a change

but when they predict the next observation they don't consider that there may be a change

so in a way when people are thinking about the future they don't consider this possibility of a change

um and in this sense uh we argue that the first definition uh the one that I call S of sh_2 is more consistent with the experimental evidence but through

the paper we take all these two versions
it appears for many of the definitions
and we do really this tedious job of
keeping all these definitions through
the whole paper but today in the talk I
only present one version of each
definition

so this was one of the examples I can
have I can present different examples of
surprise definition in the field uh with
three of them I'm going to present the
first uh

classification that we have the first
one was Channel surprise another one is
the absolute error as I said before one
can say that

given the belief that I currently have I
can make a prediction for example take
the average of these uh marginal
probability and then the difference
between these average these that was my
prediction uh and the difference between
the average and the true observations in
the actual observation is defined as a
surprise

is some sort of prediction error
and there's also this other definition
of very famous definition Bayesian
surprise that says that

I don't care that uh what is the
difference between Y_T or $\hat{Y}$ and what
is this marginal probability what I care
about is that

by observing up and observing this Y_T
plus one

how much I update my knowledge about the
world and what is the difference DKL
here stands for Kullback Divergence between

the belief before observing Y_T plus one
and after observing Y_T plus one
um it is not written in the form of Y_T
Plus One X plus one and $P_i T$ but uh in
proposing in remark 2 in the paper we
show that if PC if probability of change
was equal to zero so if there were no
change in the environment

Bayesian surprise could be written as a
difference between expected Channel
surprise and Shannon surprise itself and
funnily enough the channel surprise here
appeared with the negative sign
so um it seems like that the Bayesian
Superstition Channel surprise are really
complementary uh definitions
and uh particularly it is important to
say that we cannot really get rid of uh
the belief belief really appears remains
here so we have marginal probabilities
here but we also need the belief to
evaluate Bayesian surprise

for the case of
a volatile environment with PC greater
than zero we show that in Proposition 8
the interpretation remained the same but
um

the equations become a bit more ugly so
that's why I don't bring it here but you
can check it out
so

having these three definitions here
something that's important is that piano
surprise could be written in terms of
the marginal probability
absolute error could be written in terms
of the $\hat{Y}$ hats for an estimate of an X
observation but for Bayesian surprise we

needed the whole belief because we had to compute this expectation and that's how we make our first technical classification which is based on the dependence on the belief that this surprise definition how it depends on belief p_i

here on top we have P_i from P_i we can extract the marginal probability and from the marginal probability we can extract one estimate the first group are the observation mismatch surprise definition that only depend on the estimate for example the absolute error that I introduced

the second group are the one that only depends on the marginal probability there are one like a Shannon surprise that is minus logarithm of the marginal probability but the last group they cannot be evaluated by only knowing these two variables by knowing this distribution and the next prediction denote the whole belief like Bayesian surprise

so going back to this uh figure that I show in the introduction

so far we have seen two definitions here in this column two definitions here in this column and two definitions here in this column the first two absolute error and shine on surprise were more about the prediction so this one was saying that what is the mismatch between the prediction and observation and this one was saying that How likely or unlikely observation was while the Bayesian surprise was saying that how much

information I gained upon this
observation

okay this was a technical classification

I can go forward now and discuss the
indistinguishable conditions before

going to indistinguishable condition I

want to introduce another definition of
surprise there is this uh opinion this

idea in the field that surprising events

increased learning rates and there are

some theory for it and there's some

experiment showing that it actually

happened so then there is something

surprising the learning rate increases

but there is this um

logical question that is increasing

learning rate always good for making

better predictions or making better

knowledge of the environment or not

because uh sometimes something surprise

something can be unlikely but if it is

just outlier maybe I should not change

my learning rate I should just ignore

this observation

what we find out

um is that there is a is necessary to

have a comparison between the current

belief and the prior belief because

basically the moment that you want to

increase your learning rate upon

surprising events is when there is a

change in the environment and you want

to say that I want to get rid of the old

observation

and this base Factor surprise we

proposed it in a paper in neural

computation in 2021

and it always compared the probability

of observation under the prior belief
with under the current belief it's some
sort of hypothesis testing at each point
saying that oh whether there was a
change or not
so if this um
the nominator the probability under the
prior belief if we ignore that this is
basically a decreasing function of the
probability under the current belief
very similar to the channel surprise
so there's the less likely an event the
more surprising it is but if that event
is also unlikely under the prior belief
then it may not be surprising
and um the reasoning
become more
valid when we show in our proposition
one in the paper that actually this
definition is optimal to modulate
learning
in proposition one we show that after
observing Y^T plus one the new belief can
be written in this term which is a
trade-off between integrating new
observation into the previous belief so
take the previous belief take the new
observation combine the two make a new
belief
or throw away the old belief say that oh
there was probably a change in the
environment I should reset from scr and
start from scratch and uh put the new
observation together with the prior
belief and make a reset belief and this
trade-off is controlled by something
that we call adaptation rates here shown
by γ_{t+1}

which turned out to be an increasing function of Base Factor surprise so the higher base factors surprise the higher adaptation rate and the higher adaptation rate the more weight on the reset

basically saying that the surprise uh and this rule really comes out of the Bayesian inference is a normative rule and uh basically saying that base factors surprise uh naturally emerge or at least we Define it in a way to have uh

this decomposition of the belief in this paper we propose also a variational method that turned out to be have a very simple update rule for distribution in exponential family which we call a variational surprise minimization learning and I would like to mention here that recently it got some applications in the active inference community and people used it for model building

so I introduced the base Factor surprise as a new base uh surprise definition let's go and just overview the definitions that so far I showed the first one was absolute error surprise was in the family of observation mismatch and it was it care about how good my prediction is then there was Channel surprise saying How likely or an unlikely an observation is it was again caring about prediction but it was in the probabilistic mismatch the base Factor surprise that I just showed isn't the family of change Point

detection because as we saw it care
about whether there was a change point
or not

but it is still in the probabilistic
mismatch to price definition

[Music]

and another definition that I showed was
uh Bayesian surprise which was the
difference between uh prior previous
belief and the new belief
and is in the belief mismatch to price
definitions

um although they are in different
columns uh in different boxes one can
ask this question that is there any link
between Shannon's surprise uh here and
the base Factor surprise here because at
least there's one term that is repeated
in both definitions

that's what we show uh in the
preposition two of the paper
that Shannon's the the base Factor
surprise can be written as function of
these interesting things here that are
not Channel surprise

but very interestingly related to the
channel surprise there are difference in
Shannon's surprise so there are
different between the Shannon surprise
under the current belief and under the
prior belief

so

base Factor surprise can be written as a
function of difference in Channel
surprise so one may say that a base
Factor surprise has a interpretation
like a relative surprise and in that
sense it always compared the prior

belief with the current belief and
that's why it's good for learning
as a consequence in corollary one you
can argue that optimal surprise
modulation of learning can also be
written in terms of these differences
instead of the base factors to price
but the most fun thing is that in
corollary 2 we can
we show that if the marginal probability
under the prior belief is uniform so if
basically it means that if this term is
constant then there are strictly
increasing mappings between base Factor
surprise and two definition of Shannon
surprise
and strictly increasing mappings
basically means that these are
one-to-one uh
function of each other so they're
invertible and if I have one of them I
can find the other two so in a way
they're indistinguishable
and that's one of the big results of our
paper that's
the uh I identify the conditions under
which different definitions are
indistinguishable
so here in this figure you see different
definitions they are connected to each
other with lines each line corresponds
to one of these conditions that if this
condition is satisfied then these two
definitions are indistinguishable
and uh C2 here means that
this condition is proved under corollary
two in the paper
why this uh indistinguishability

conditions are important
um the first thing is for experiment
design for example let's take the um
Oddball test that I introduced before
usually an oddball task it is assumed
that uh marginal distribution under the
prior belief is flat because there's no
preference for the uh at the beginning
there is no preference between the blue
disk and the Red Square as a result
basically means that in this kind of
experiment all these five definitions
that I have here are indistinguishable
so I cannot do the Oddball task and
expect to see a difference between these
five definitions
in addition some people argue that even
the prior belief is flat and if is that
the case then these two definitions are
indistinguishable
even more because Oddball task is a
categorical task it is one of the
special cases that we identify which
basically means that
seven of these definitions are
indistinguishable so it's quite
important that if we want to make
experiment to distinguish some of these
uh surprise definitions to study
different definitions of surprise we
should be careful to
not have experiment where these are
indistinguishable and what we do is that
we take a series of papers in the field
that have experiments
where they study different definitions
of surprise and we identified which of
these conditions are satisfied or not in

these papers

the second thing that these industry
increasability condition can be
important for is for computational
modeling there is this page this new
preprint from orlab and Via archive
they're um

the authors show that they design a
spiking neural network that try to learn
in a volatile environment and modulate
its learning rate with the notion of
surprise

in the setting that they analyze we can
show that these three lines the
condition corresponding to these three
lines are satisfied so basically these
definitions of surprise are equivalent
what is uh

why this is interesting for us because
this one is optimal for learning is the
base Factor surprise that I showed
before while this one is pretty easy to
implement while base Factor surprise was
a function of probabilities well this
one is the observation space and that's
actually the fun thing that what they do
here is they make a spiking Network that
computes something related to this
observation mismatch surprise
definitions while there are some uh
optimal justification for why these um
surprise definition can be helpful for
learning

so putting together so far I proposed
the I presented the technical
classification that we proposed and
indistinguishability condition that we
found out

and now I'm going to discuss the
taxonomy

uh

I had 40 minutes so I still have time
right I can

so for your taxonomy I need to introduce
another definition I promise is the last
definition that I uh introduced today

it's um the notion of confidence

correction I want to introduce that and
there is this idea that higher

confidence higher confidence must lead

to higher feeling of surprise so the

more confident you are about your

predictions the more surprise you should
feel

energy at all uh in this paper proposed
this definition

uh it may look a bit strange at the

moment I would demystify it in a bit

it's the difference between the current

belief and after observing this

observation

how much uh what would be my belief if I

would cons if I would throw out all the

previous observation and make a new

belief so it's a next belief but

with the assumption that there was a

reset just for a comparison between the

confidence corrected surprise and

Bayesian surprise in the base and

surprise I had a difference between the

current belief and the next actual

updated belief

but for the confidence corrected

surprise is the current belief with the

reset updated belief

uh the reason that is interesting is

that in proposition 9 of the paper we show that this definition can be written in this way where the first name is different in Shannon surprise the second term is kind of difference is based on surprise and the third term is a notion of confidence

so this one looks like a relative surprise as I argued before it's good for uh modulation of learning and that was actually the original plan of Heritage at all to find the surprise definition that is good for surprise modulation

and the last term is a notion of confidence and in these pre-print online we show in some situation uh it seems more intuitive to have some confidence correction in the definition but I don't want at the moment to um enter that discussion I just want to introduce that there are this class of definitions with a confidence correction an overview of all definitions so far the absolute error but the difference between observation and prediction in this class

Shannon's surprise how unlikely something is base Factor surprise uh how unlikely something is compared to a baseline or prior belief

the Asian surprise how um how much I gain information upon observing something and confidence corrected surprise I should include the explicit notion of confidence

having these five definitions now we saw one definition per row and one

definition or at least one definition

per column

now I can introduce our taxonomy

on the on the screen now you see in a

seemingly arbitrary order these

definitions that I had here

and we propose to label these different

definitions based on the quantity they

measure

this group that's what we call a

prediction surprise they care about how

good your prediction is about the next

observation

this group where information gains

surprise they measure how much you learn

based on a new observation and these

groups for change Point detection

surprise that were some sort of relative

measures of surprise

and we had this uh overlapping labeling

that no matter where there is a

prediction or change Point detection we

can make the effect stronger or we can

correct the effects by adding an

explicit term for uh confidence

one can I ask that okay so far I talk

about vendor indistinguishable and put

them in different category one can ask

how different are these definition in

practice because if I um

pick a random Oddball task from the

field look at the data all these

definitions look quite correlated

and uh one may argue that okay

does it really matter to have different

definitions

and um

it is actually possible to design

experiments here I'm showing one example
I'm not going to discuss that what is
the experiment to be presented in this
preprints on the x-axis you see time on
the y-axis average surprise and four
definition from these four different
categories and here is a change point in
the environment and we see after change
point they have a completely different
Behavior one goes down one has Spike one
goes down and go up again and down again
so there are possibilities to make
Theory or even experiments that
different definitions of surprise can
have very different behavior and
different predictions
another example is um
in this preprint the focus and the role
of different surprise definitions and uh
curiosity or even exploration that how
differently they can drive exploration
and um there are also many other works
and that are not from our lab and they
work and they focus on the physiological
signatures of different definition of
surprise particularly this one is in
Pillars computational biology I found it
quite impressive they found signatures
in EEG for all three groups of
prediction surprise confidence
correction surprise and Information Gain
surprise so a fixed tax a fixed
experiment looking at the EEG they could
see that all these three have their own
correlates or signatures in the EG
signal
so I introduced the mathematical
framework

and formally discussed many definitions
of surprise we had this technical
classification and distinguishability
conditions and the taxonomy
take on messages one surprise in
different experiments refer to different
definitions

these different definitions can be
classified based on on their dependence
on the belief on the subject's belief
and the quantity they measure which was
the taxonomy or the labeling we proposed
and the third message under specific
condition some of these definitions are
indistinguishable and I argued that it
is useful for experiment design and also
useful for computational modeling
so putting everything together back to
this figure that I showed at the
beginning of the presentation
our proposition is to replace or at
least order these different signals or
definitions

in a structured way that
um

it

we can now uh if you put it in this
structured way let me rephrase it in
this way that then we can talk about
that oh for example for exploration
reaches these different domains is more
relevant is uh the belief is my surprise
definitions that are more relevant to
acceleration or the observation is my
surprise definition so we are not
talking about only specific definitions
but more like categories
um in addition to this uh surprise

related signal or these surprise
definitions we argue that there's also a
need for uh novelty signals or novelty
definitions
and um

we argue in this paper that novelty and
surprise are fundamentally different so
everything that I said today would not
uh stand for
novelty

so I would like to thank my um
collaborators for this joint work or
very cool lab in lausanne and you for
your attention thank you very much
and you all right just getting things
back awesome

I will restore our videos well thanks
for that awesome talk if anyone has a uh
question in the uh live chat please feel
free otherwise Ali if you'd like you can
re-enable your video and feel free to
give a first comment or I'm happy to
give a comment but you can go first uh
yeah sure but for some reason I can't
activate my camera here

uh

uh let me check it again okay

uh so thanks a lot

uh first of all I congratulate you for
writing and your of course your
co-authors for writing such a
fascinating paper

thank you I truly enjoyed it

uh well I have some

um comments if I may and

um also I have uh maybe a couple of
questions

uh

your work actually these kinds of classification work or the taxonomy is in my opinion is really a significant contribution to the whole area of psychological or especially in cognitive Sciences because as Muhammad Ali khalidi's recent book The cognitive ontology claims this kind of taxonomic approaches has been long debated among the neuroscientists psychologists and cognitive scientists especially uh about the difference between cognitive kinds and natural kinds and whether or how they map unto each other because as you also mentioned we have a folk psychological notion of surprise or uh let's say cognitive uh kind of surprise but we're not always clear about how to translate that into precise mathematical terms so I remember in active inference book by par pazulo and Kristen there's about a paragraph or so distinguishing between the folk psychological notion of surprise and the statistical surprise or surprisal functions corresponding to various distributions so basically as I see it your paper is a kind of expansion on that single paragraph to consolidate all the different conceptions of surprises into a whole unified framework we were talking the other day about the importance and significance of the recent move toward applied category Theory as a kind of as I like to call it the conceptual housekeeping or better still a kind of cartography of the

knowledge trained so this this kind of
research

in my opinion can be seen in congruence
with this recent move

toward as Toby Sinclair Smith would call
well to do a kind of well-typed science
to be

precise and clear about our definitions

uh and uh I also wanted to uh kind of
getting back to that uh distinction

between the cognitive kinds and natural
kinds

um there are different taxonomic

approaches in cognitive science

addressing these problems uh but uh as I

understand it uh your work here is more

in line with the revisionism approach to

do these taxonomic uh these

classification approaches which says

that which which basically says the

current cognitive taxonomy is not

precise enough to adequately address uh

the problem of

uh distinguishing between cognitive

kinds and natural kinds so we need a

more precise formalism to do that

reminds me of uh Dalton's active advil's

notion of blanket index introduced in

his paper a week a mark of blankets

because before

before the publication of that paper

there was much confusion and ambiguities

about the exact mathematical definition

of the markup blanket or to put it in

other words to make it more General and

more precise what do we mean exactly by

Mark of blanket so my question is

as I see it uh so far you have

a somehow mapped these kinds of
different surprises

in a kind of discrete space as you
showed in one of your tables
so do you see it as a feasible move
forward

or let's say a feasible move toward more
General conception of this framework by
conceiving

this unified framework as as a
two-dimensional continuous
or n dimensional continuous manifold of
possible definitions of surprises and
maybe even Define an index to navigate
this continuous manifold space because
in that case

obviously you could the problem possibly
avoid

um those kind of blank

blank spaces you have on your tables and
maybe even we can gain a kind of a much
deeper insight into how exactly those
different kinds of surprises relate to
each other in a much more continuous
sense rather than the discrete stance so
do you think this kind of move even
feasible from your viewpoint

now that's a very good point uh thanks a
lot for the for all the remarks I
enjoyed it very much

um

the way that I Define surprise in a
general sense was this function or
functional that get the Q observation
and the belief and map it to a real
value number so all the types of
functional in this space can be seen as
a measure of surprise so the whole Space

is quite big and uh
continuous uh so in principle yes of
course we can do that but the space is
too big actually we need to put some
constraint on it so I think for future
work if one want to go to that direction
uh one should say that what are the
properties from a surprise definition
that I would like to have so put some
constraint on this space and then in
some sense uh parameterize this this
space to have some interpretation
because at the moment I'm saying that
okay let's look at this part of this
space that these definitions were there
people are already talking about these
definitions and I would call these
definitions for example Information Gain
surprise definitions so maybe one can
really uh formalize all the definitions
that it is possible to call
because okay I only talk about these two
particular definitions with Kullback
Divergence if you replace scale
Divergence with any measure of uh
distance on the space of uh
probabilities we end up with another
information in Surprise definition so I
think it is feasible in principle but I
think it's a bit of a tedious job and uh
whether how useful it's going to be
is not clear for me because we already
have many definitions and as I try to in
a way say
what justify definition is at the end of
the day how useful it is to explain
experimental evidence and help us think
about cognition

Justified true belief distribution

um

thank you irreza want to add a general
comment and then bring us back to active
inference so

in exploring the active inference

ontology we've often found this tension

where a given term would have one
definition very folk psychological very
everyday

belief surprise observation sense action

these are some of the most common words

in any language and then your work

started to dissect or characterize due

taxonomics on that Iceberg and show that

the pointer of a given word surprise in

this case can actually host uh diversity

of technical definitions with some

subtle

contradictions or compatibilities and

areas of indistinguishability so I think

it really

um raises our attention to the Horizon

and shows that for a lot of terms that

people again use in a day-to-day context

there may be multiple definitions and

there's many implications like there

isn't a single way to model action or

belief or surprise

there are different flavors and even

within a flavor there's going to be

different ways to deploy it in a given

model or script or modeling context so I

think the formal

coherences that you're providing are

going to go a long way to ensuring that

given that kind of pluralism in modeling

and statistics which has always been the

case that we do have connections that are robust to enable comparison like you did with different empirical setups whether the experimenters understood sh1 versus sh2

we can

um post hoc essentially as peer review annotate or augment the definitions re-partition different ways that they analyzed data and so I think that really opens up a lot of directions something that you had on several slides but didn't really go into that much was F star

so could you add a little bit on free energy

what is free energy what's free about it what's energetic about it why do we use it to

help us engage in perception cognition action in the free energy principle and active inference

thanks a lot for the question

um the way that I see free energy in the context of what I discuss about is as long as I talk about exact Bayesian inference and really have my exact update rule for the uh for the belief going from the previous observation and previous belief to the exact next belief uh free energy doesn't play much of a role free energy plays a role as soon as I want to talk about approximate inference and go to the variational setting and in that case basically free energy defines

um some loss between my approximation and the exact true belief

and one can show that this is the upper
Bound for the Shannon surprise that they
Define
and the difference between the the free
energy and these uh final surprise is
how good my approximation is and that's
why if I do exact inference the minimize
for energy is equal to the channel
surprise so in that sense I cannot
really distinguish minimize free energy
from Channel surprise but if I work in
the uh approximate inference setting
then minimize free energy is the best
approximation of Shannon surprise that I
can get
by
like this in that framework
yeah
awesome
just to kind of restate that because
it's actually one of the fundamental
aspects of the approach taken in active
inference if the true state of the world
is a gaussian and we're doing exact
inference with the gaussian prior
we will seek to minimize our Shannon's
surprise
we could
if the world is not truly a gaussian but
it's some other unstated or unstatable
distribution but we choose to do
variational approximate Bayesian
inference with a gaussian prior then
free energy tracks us doing strictly as
well as we could
given our choice of Prior family and so
it operationalizes with approximate
Bayesian computational techniques

these formal definitions which may or may not be tractable in large State spaces or with large amounts of data so it's a super important point and it's almost like there's that Iceberg of surprise definitions and then free energy

ports us into the approximate which enables us to to get some tractability in settings where inexact Bayesian approach is not plausible okay two things first is even more than that so uh even no matter what is world

I may assume that world is a as a quotient distribution let's say but there's some dependency and I still even though I'm aware of this dependency I can assume that there's no dependency and do that in my uh inference so when I think about the world it is actually true I believe that there are dependency but they still do approximate inference so I think when we do variational inference we may not necessarily care that what is the true distribution we assume the distribution has this form the yet we want to approximate it with something simpler uh this is the first part and the second part

all the definitions of surprise that I talked about can be applied to this approximate uh setting as well so I can use the belief approximate belief that I found with variational uh inference and just imported in any of the definition that I

had and come up with some definitions of surprise I think what I want to emphasize on is that variational free energy and minimize free energy is one of this particular one but all of them are still applicable in the approximate variational setting

thank you for the correction like you mentioned variational free energy and there is expected free energy about future but also people have proposed variants like the free energy of the expected future and so it speaks to categories of functionals that do different computations and there isn't just one specific functional form they might highlight different aspects as your plot of surprisal through time let's just say it was some stimuli and I think that leads to

um a next question which is if we're interested in some biological phenomena how does this work

help shape our experimental design statistical power analysis discussion section like how do the thousands of papers surely using surprise in a qualitative or a quantitative way

how do we translate this work to increase the rigor and interoperability of those studies

that's a very good question uh and that's somehow the ideal uh place that I want or paper to be used that uh it makes a framework uh that we can think in a very formal way that if I wanted to design an

experiment to distinguish these two
different Notions

what would my generated what was the
necessary necessary features that my
generative model need to have and how
can I uh make the realization of that
generative model in an experimental
paradigm what we do for example in the
preprint uh is not in the main paper
using a preprint that is online it's a
and bio archive with the same set of
authors there we try to find situations
that for example it's a some election
situation that each week before the
election is on the news media tell you
that uh this party is going to win with
this probability then you see the
results and we try to make some
assumption that what are the features of
these uh experiments and what we see is
that oh actually different definitions
make very different predictions so I
cannot give a systematic approach that
oh okay this is the algorithm that if
you want to design an experiment just go
one by one these steps but I want to say
that this framework gives us the tools
to think about before doing the
experiment how to design it to have the
maximal effect in many of these also
indistinguishability conditions I say
for example that if something is flat
these two definitions are
indistinguishable the thing is that if
that distribution is not flat the two
are distinguishable but the effects that
is very small if it is close to flat so
the best thing is that if I want to make

two uh definition of surprise is
indisting
distinguishable in an experiment then
it's best to look at that condition
and make the opposite of that condition
in my experiment and in that way I make
the FX size as big as possible in some
sense it's a very um
non-rigorous way of talking about uh the
theory but just intuitively that's how
we design those experiments and could
convince or experimental colleague at
the moment to run the experiment so we
are doing that
awesome yeah there's a lot of ways that
we could imagine
pre-registration of studies and just the
development of templates and patterns
for statistical analyzes so that
different kinds of surprise could be
highlighted and we could say if the
change point is going to be like this
then you will be able to distinguish
with this power these kinds of outcomes
versus another um Ali do you want to
question or
yeah I actually this discussion reminds
me of uh Terence Deacon's uh famous
paper uh what is missing from theories
of information
uh because uh in that paper he also
claims that one of the misunderstandings
about uh all the uh theories of
information or all the relevant uh
theories for that matter is the
assumption that
in order for any information to have
a kind of

for any content of information to have
any real world consequences
it must have some substantial properties
and must of correspond to some real
world situations or something present as
an extrinsic entity in some form another
but as he shows in that paper we can
regard the information whether as a kind
of probability distribution or any kind
of other theoretical construct
as

exactly conveying the kind of
information that uh we want we wanted to
explore in a much real sense and it's
not just pure data or pure uh
abstract information for that
so for instance uh Shannon's analysis of
information of course is based on the
constraint and the entropy but on the
other hand the capacity to convey that
information is uh it depends on the
relation to something that is
specifically absent or not produced so
this absence or presence of the entities
that the information is about that
entity or so-called the aboutness of the
information is

I heard the aboutness of the information
but we'll pause and see if he'll rejoin
otherwise we can continue
okay

I'll just kind of
do you want to provide a thought on
Shannon or just uh yeah that's actually
very um important point that uh Ali was
about to talk about uh which is
in all these uh
definition that I propose we didn't

really care about uh what is uh why or
what is X we were working in these very
abstract States and a very abstract
framework one may say that what if I
don't care about these observations at
all and I still may have some
predictions and uh
so this aboutness that you were talking
about is quite important and whether I
no matter how confident I am or what is
my prediction how much I care about that
problem is quite important and how
surprised I am if I don't care about it
at all then I may not feel surprised at
all
um so that's something that's really
missing from the whole framework that
how we can model the
um
the the other things that have let's say
semantic because at the moment uh all
the framework is semantic free
uh how how can we include some uh
semantic into this framework we can talk
about uh
then we have another dimension to talk
about surprise
yes
and some hope that with
Progressive
syntax schemes
we can partition that semantics into
other surprise definitions
but the bigger question is what are we
surprised about and I think that
connects to these other cognitive
phenomena slash
complex word complexes like attention

confidence salience

um and I mean just to kind of put a point on that like when surprise is being used as an imperative for optimization and inference or merely as a signal that plays into action selection

your robot or Aunt or person or software is gonna do different things depending on the surprise

formalisms used

and knowing many flavors of formalism

it would be awesome if we could have a software package that computes 18 different numbers

but then what would we do we could choose one we could blend them we could make summary statistics we could say that we're in zone a where these five are aligned but this one isn't and then what would we do with that action how surprised would we be by that

location and phase space

so it's like a very interesting

top-down

situation independent phrasing of those relationships but in the implementation it's always going to come down to what the modeler chooses

there just isn't a general answer to be found in the specifics

one other thought

or area to explore is

you mentioned the Bayes factor and its role in learning so could you speak to optimal Bayesian inference and optimal learning

what do you mean sorry and you like

Bayes optimal inference

okay that's what I meant that base

Factor surprise appear in base optimal

learning and also appear in the

variational uh Bayesian update Rule and

that was actually what we found cool

about it because neuroscientists

particularly people talking about the

synaptic plasticity talked about

surprise and how neuromodulators

influence synaptic plasticity

and we saw that if you're looking at the

Bayesian framework we can find that such

normative modulation based on surprise

exist

in how the belief is updated and it is

funny that the same thing appear in lots

of approximate inference scheme as well

and it seems necessary to use such a

surprise measures to modulate learning

as we learn and there are lots of

experimental evidence for it

um so

what I want to say is not necessarily

only about the exact base and inference

but it is something that appeared there

and is useful for also other kind of uh

other approximation to exact inference

so what does optimal learning in this

case mean

um

optimal learning

means to find the exact posterior

given the model for those parameters

because if the model is true if I

believe that this is the true model

the optic all the information that I can

have about the hidden variable

is in the posterior distribution of that variable given the observations and basically what we mean by learning is to

um updating this uh posterior distribution from one time to another so as we go on and add more and more data points how we track this distribution over time and how we update our belief and by Optimal learning we mean the exact up Bayesian update from time T to $T + 1$. but this could be done also with variational inference

but the cool thing was that also Bank variational inference would benefit from such modulation

so one thought on that in a little bit of a

a meta science area is Bayesian statistics is becoming more prevalent and one way of reporting the effect of a treatment would be like with a base factor to present the relative evidence for one model versus another and something that that plays a functional similarity with but of course a technical difference is a p-value and so in the presentation I thought about how the base Factor was associated with learning as well as we could

representing the evidence for two different possibilities in a ratio whereas a p-value especially when we put um a discontinuity 0.05 is like a change point

it's like we're sampling from the experimental outcomes and eggs are

healthy eggs are healthy eggs are
healthy now there's a difference between
the groups
so that puts the task of Science and
learning into a change Point detection
framework
where oh now we're in the world where
this is the case
versus the sort of unrolling Bayesian
approach
where we get
some experimental results
that are all over the posterior
but merely drawing results from a
potentially quite dispersed
posterior
doesn't mean that we've entered any
change point with respect to the
efficacy of a given treatment
I'm trying to uh
to connect what you said about science
in general to what you were thinking
just as a point that the base Factor
surprise we really took the word base
factor from the field of statistics for
the nature of hypothesis testing that
exists in these kind of change Point
detection because at each point you want
to see that oh okay was there really a
change in the environment or not and uh
for example for this uh
weather forecast
in the pandemic I really had this before
knowing that uh this is really a
phenomenon and weather forecasts get
less accurate for a couple of weeks I
was confused that okay what's going on
why my uh metal Swiss is not working uh

well anymore

so it's always some sort of hypothesis

testing that okay am I living in the

same world that it was yesterday

or something changed from yesterday so

there is some hypothesis testing and in

the Bayesian setting it's done by

comparing two models to base Factor but

we can also have even in the Bayesian

setting if we go again towards

approximate Bayesian setting and

sampling like the particle filtering we

sample this change pouring so at some

point we make decisions that uh oh this

is the probability of for example

there's a change or not at some point I

should decide I would say yeah there is

a change and from today onward I behave

as if something changed yesterday

and uh in a frequentis approach

is a bit difficult for me to include

things like p-value in the framework

that I was talking about because the

nature of p-value is in talking about

probability of things more extreme than

things that already happened while in

this Bayesian setting we are talking

about this particular observation I

don't care about the values higher than

observation lower than observation

different from I I care about only this

observation well in p-value we compute

the probability of observation more

extreme than this one

and um

and I don't see how it can easily fit in

the framework but if I want to find the

closest measure to say that it's a bit

similar to p-value I would say it's something like Shannon's surprise because it doesn't have the nature of comparison you just say that under the current belief which let's say that the null hypothesis is the current hypothesis how unlikely is the new observation yes one could certainly apply a change Point base Factor over three means we publish the paper base Factor under three is no

evidence and there's definitely some subtlety there with what the P statistic and other empirically calculated statistics

tell us

I'll read one question from the chat

then Ali so Dave writes

have folks working on these problems modeled the situation where counter evidence has no impact on confidence delusion and cognitive dissonance are terms that come into that discussion

I'm sorry can you read again because I opened the page to read it myself as well but then all good have folks

working on these problems model the situation where counter evidence has no impact on confidence delusion and cognitive dissonance are terms that come into that discussion

that counter evidence doesn't have effect and confidence

that was yes potentially where um surprising beliefs are continually entertained

how could we model a situation where persistence

uh cognitive dissonances or delusive
beliefs are stabilized

okay

about cognitive doesn't

when I talk about hypothesis testing in
a Bayesian setting the fun thing is that
all hypotheses live at the same time you
just change the weight over them and
some of these hypothesis are in
disagreement with each other
so this

counterfactual assumptions that oh what
if there was a change a while ago or
what if uh

oh wait

no the previous observations are fixed

I have this uh

I have this imagination only in the
unobserved States and when I talk about
uh different scenarios I talk about
unobserved States

what

um

I think the question was about
if I want to think about counterfactuals
then I should imagine that how would the
world be if the observation was
different

and that at the moment is not included
in the framework

but one could think about it if we would
uh integrate the framework it's a bit of
causal inference

and in those settings we have this
counteractual and talking about now I
don't only condition in the past I
imagine that some of the observation is
very different and now I do my inference

but I'm not sure if I got the question
and if my answer was relevant I hope it
is uh
related
awesome and on that
um response Ian wrote cognitive
dissonance I think there's something to
do with having different beliefs at
different levels of consciousness so yes
these
taxonomies were presented at a single
level model as presented and with nested
models we might have
um different interactions amongst
surprise definitions maybe even
different surprise definitions playing
out within a nested Bayesian
architecture
maybe one type of surprise is more
amenable to Edge detection in the retina
and then in a different part of the
generative model there's another Opera
operationalization of surprise in a
decision making or in a risk aversive
setting
of course that's uh definitely the case
that different
level of processing may work with
different levels of surprise and
um maybe I don't know uh belief
mismatched price definitions are not
relevant for the primary uh part of
Cortex at all
but
if that's the case yeah I understand but
uh maybe I misunderstood what they mean
by cognitive
Ollie any more thoughts or questions

maybe we can each have one more question

or

contribution

yeah actually uh first of all I

apologize I got disconnected in the

middle of my question I don't know uh

how much it was fine I could yeah uh all

right so one thing about these kind of

nested hierarchies

um is that in the interactivist

framework uh they usually uh Define

different kinds of information as

different levels of emergence in a kind

of nested hierarchy uh the biggest of

all is the Shannon information which

deals with the kind of medium capacity

of

um conveying the information and then uh

on another emergence level is uh the

referential information which is about

the aboutness

and

um at the highest emergence level we

have the significant information or uh

as they would like to call the

usefulness or the semantic content of

the information

as they say so my question is that uh

this kind of tax taxonomic approach that

you uh

provided in this paper uh how can we map

this taxonomic system or even whether we

can map this kind of taxonomic system

into this kind of nested hierarchies of

different information because obviously

uh we can probably even say that the

nature of these different kinds of

informations are different right

um my short answer would be
the table that I showed
if we want to account for the things
that you discussed we should make a few
of those tables so we should add another
dimension so I don't think it's really
possible to map this particular
definitions to those different uh
categories
uh

and for the long answer

I give again another short answer which
is basically we need to make a new
framework to really include all these
semantics or current probabilistic
framework the one that I presented today
doesn't have the capacity to uh account
for that

yeah that's why I mentioned that in
manifold thing I said my first question
because I was thinking of something
similar because uh maybe uh in order to
generalize this framework we would need
a lot more Dimensions than uh just the
two Dimensions that we have already
provided in the paper so uh I don't know
if as you say if it would be
useful for any uh future research or
whether it would be any kind of
attractable uh computational framework
but at least in my opinion or from
purely theoretical point of view
maybe it's worth investigating to see
how these kinds of different Frameworks
uh integrate into a much more
generalized way of thinking about
information and surprise but as I said
your paper is a very important and

significant contribution in that regard
so in near closing the
taxonomy notion it's a classification
scheme and just as in Biology
early Natural History observations of
different things led to taxonomics and
the classification of different things
and on through phylogenetics and
understanding the the temporal
relationships so today we have a
taxonomy for surprise that we didn't
have
a year ago
what can be contributed by your group
and by others in this emerging ecology
of formalisms
if we have the taxonomy today just a
sketch napkin sketch two by two grid we
sorted the birds that people saw into
roughly these categories they don't even
know that it was this bird but now we
can disambiguate it was that bird
so
where will you be heading
in this space and where do you think
there are areas for people to contribute
to this living area
that's a very good point a very good
question uh there is this slide that I
had these different four domains of uh
learning exploration memory and
physiological signals that people talk
about surprise in neuroscience and
psychology in the middle these different
definitions I think
if you want to go beyond the taxonomy
we should try to do more computational
modeling and see that which of these

definitions which part of these two
degrees is related to which of these uh
function so basically now I'm
just proposing some tools and a
framework which can be used later on for
computational modeling so all these
definitions can be useful or not for
some modeler to explain exploration in
some task or learning in some other
tasks or segmentation in some task so
going to be the next step is really
finding out where and what function this
definition makes sense
yes you brought up the novelty curiosity
learning and surprise is woven sometimes
technically or not into those
discussions
but it's not all definitions a surprise
for all definitions of learning salience
attention novelty so disambiguating that
space is very important
Ollie any final remarks
uh I would like to thank you again for
inviting me and
thanks a lot for all the discussion and
nice question oh awesome it's a great
honor and learning experience for us so
you're always welcome back till next
time
you're really appreciate it thanks any
reason thank you have a nice day bye-bye
very well bye